

R307S Fingerprint Sensor - Complete Overview

Introduction

The R307S is an optical fingerprint sensor module widely used in embedded systems, attendance systems, smart locks, access control devices, and biometric authentication projects. It contains an optical imaging sensor, DSP processor, flash memory, and UART communication interface.

Key Features

- Optical fingerprint sensing
- Internal fingerprint processing and matching
- UART (TTL) communication
- Stores up to approximately 1000 fingerprint templates
- Fast identification and verification
- Suitable for ESP32, STM32, Arduino, and other microcontrollers

Pin Description

VCC: Power supply (typically 5V)

GND: Ground

TX: Serial data output from sensor

RX: Serial data input to sensor

TOUCH: Finger detection signal

3.3V: Auxiliary/internal voltage pin (module dependent)

Working Principle

When a finger is placed on the sensor, an optical image is captured. The internal processor extracts fingerprint features and creates a template. During authentication, a newly captured fingerprint is compared against stored templates.

Enrollment Process

1. Enter a fingerprint ID.
2. Place finger on the sensor.
3. Remove finger.
4. Place the same finger again.
5. The sensor creates a model and stores it in flash memory.

Verification Process

The sensor captures a live fingerprint, generates a template, and compares it with stored templates. If a match is found, the corresponding ID is returned.

ESP32 Interface

R307S VCC -> ESP32 VIN (5V)

R307S GND -> ESP32 GND

R307S TX -> ESP32 GPIO16 (RX2)

R307S RX -> ESP32 GPIO17 (TX2)

Applications

Attendance systems, smart door locks, biometric authentication, access control, embedded security systems, and IoT-based identity verification.

Project Example

Fingerprint Authentication System using ESP32 and R307S:

Authorized fingerprint -> Green LED ON

Unauthorized fingerprint -> Red LED ON